ModelIt!

Student Activity Pack

The Opioid Epidemic: When Medicine Becomes the Disease

Modeling Neurochemical Feedback Loops in Addiction

Grade 09L2 | NGSS HS-LS1-2, HS-ETS1-4 | Cause and Effect: Mechanism and Prediction

STUDENT NAME _____ DATE _____

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do you think some people become addicted to prescription painkillers while others do not?

2 What do you think happens in the brain when someone takes an opioid repeatedly?

3 Draw a diagram showing how a person goes from taking medicine for pain to being unable to stop.

4 What does tolerance mean in the context of drug use, and why is it dangerous?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Prescription Rate
- Tolerance Development
- Addiction Level
- Dopamine Receptor Sensitivity
- Social Network Exposure
- Pain Threshold
- Withdrawal Severity

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Short-Term Use

Moderate prescription, 2-week duration

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Chronic Escalation

High prescription maintained over months

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Network Effects

One prescription feeds social exposure

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Naloxone Availability — Accessibility of opioid overdose reversal medication, which saves lives but does not address underlying addiction

Medication-Assisted Treatment Access — Availability of evidence-based treatments like methadone and buprenorphine

Illicit Fentanyl Contamination — Presence of ultra-potent synthetic fentanyl in the drug supply, dramatically increasing overdose risk

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does a pill prescribed by a doctor become the deadliest drug epidemic in American history?
Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that addiction is a neurochemical process rather than a moral failing?**

2. **At what point in the feedback loop would intervention be most effective and why?**

3. **How do social network effects transform an individual problem into a community epidemic?**

4. **What are the ethical implications of modeling addiction?**

5. **How does opioid-induced hyperalgesia create a particularly cruel feedback loop?**

8 STEM Challenge: Design an Opioid Crisis Intervention Dashboard

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YOUR ENGINEERING MISSION

Design a data-driven intervention system that identifies at-risk communities and recommends targeted interventions at different crisis stages.

THE SCENARIO

A state health department has $50 million to combat the opioid crisis. Your team must design a dashboard that identifies which communities need which interventions based on real-time data.

Engineering Design Thinking

1. WHAT DATA SOURCES WOULD FEED YOUR DASHBOARD?
2. HOW WOULD YOU VISUALIZE RISK LEVELS FOR DECISION-MAKERS?
3. HOW WOULD YOU ACCOUNT FOR THE LAG BETWEEN PRESCRIPTION CHANGES AND OVERDOSE OUTCOMES?
4. HOW WOULD YOU MEASURE WHETHER YOUR INTERVENTION IS WORKING?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Epidemiologists study disease spread through populations, tracking prescription patterns and intervention effectiveness, earning $60,000-$120,000/year at CDC, state health departments, and universities.